

WEB ENGINEERING

ALUMNI REGISTRATION SYSTEM

HTML5 + CSS3 • PHP • MySQL/MariaDB • XAMPP

LAB DURATION 150 Minutes | Platform: Windows | Localhost development

Build a complete alumni registration workflow: form → PHP backend → database → alumni directory.

Item	Specification
Frontend	HTML5 + CSS3
Backend	PHP
Database	MySQL / MariaDB
Server	Apache via XAMPP
Database Tool	phpMyAdmin
Editor	VS Code / Notepad++
Difficulty	Beginner → Intermediate

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1. Lab Overview

In this laboratory, students develop a basic Alumni Registration System for a university or college. An alumnus/alumna enters personal, academic, employment and contact information. The HTML form sends the data to a PHP backend, which validates the input, securely hashes the password and stores the record in MySQL.

REAL-WORLD CONTEXT This module can become the foundation of a larger Alumni Management System with login, profiles, admin approval, search, CRUD operations, events and an alumni directory.

2. Learning Objectives

- Create a professional HTML5 alumni registration form.
- Use CSS to structure and style a web form.
- Submit form data with HTTP POST.
- Receive and validate form data using PHP.
- Connect PHP to MySQL/MariaDB using mysqli.
- Insert records using prepared statements.
- Hash passwords securely with password_hash().
- Retrieve and display alumni records.
- Run and test the application locally using XAMPP.
- Understand the frontend → backend → database workflow.

3. System Architecture

```

ALUMNI
|
v
HTML/CSS Registration Form
|
| POST
v
PHP Backend (register.php)
|
v
Database Connection (db.php)
|
v
MySQL/MariaDB (alumni_system)
|
v
Alumni Directory (alumni.php)
  
```

4. Required Software

Software	Purpose
Windows	Operating system
XAMPP	Local development environment
Apache	Web server / PHP execution
MySQL/MariaDB	Database server
phpMyAdmin	Database administration
VS Code / Notepad++	Code editor

Chrome / Edge / Firefox	Application testing
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5. 150-Minute Schedule

Activity	Time
Introduction to Alumni System	10 min
XAMPP configuration	10 min
Create project directory	5 min
Create MySQL database	15 min
Create alumni table	10 min
Create database connection	10 min
Build HTML registration form	25 min
Apply CSS styling	10 min
Develop PHP backend	25 min
Test registration	10 min
Display alumni records	10 min
TOTAL	150 min

6. Project Setup in XAMPP

Start Apache and MySQL in the XAMPP Control Panel. Then create the following folder:

```
C:\xampp\htdocs\alumni-system
```

Final project structure:

```
alumni-system/
├── index.html
├── register.php
├── db.php
├── alumni.php
└── style.css
```

IMPORTANT Do not open PHP files by double-clicking them. Use `http://localhost/...` so Apache can execute PHP.

7. Create the MySQL Database

Open `http://localhost/phpmyadmin`, create a database named `alumni_system`, select it, open the SQL tab and execute:

```
CREATE DATABASE alumni_system;

CREATE TABLE alumni (
    id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(20) NOT NULL,
    gender VARCHAR(20) NOT NULL,
    date_of_birth DATE NOT NULL,
    student_id VARCHAR(50) NOT NULL,
    degree VARCHAR(100) NOT NULL,
    department VARCHAR(100) NOT NULL,
    graduation_year YEAR NOT NULL,
```

```

current_job VARCHAR(150),
company VARCHAR(150),
city VARCHAR(100) NOT NULL,
country VARCHAR(100) NOT NULL,
address TEXT,
password VARCHAR(255) NOT NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

Field	Purpose
id	Unique alumni record ID
full_name	Alumni full name
email	Unique email address
phone	Contact number
gender	Gender
date_of_birth	Date of birth
student_id	University/student registration number
degree	Degree/program
department	Academic department
graduation_year	Year of graduation
current_job	Current designation
company	Current organization
city / country	Current location
address	Current address
password	Secure password hash
created_at	Registration timestamp

8. Database Connection — db.php

Create db.php inside the project folder:

```

<?php

$host = "localhost";
$username = "root";
$password = "";
$database = "alumni_system";

$conn = new mysqli(
    $host,
    $username,
    $password,
    $database
);

if ($conn->connect_error) {
    die("Database connection failed: "
        . $conn->connect_error);
}

$conn->set_charset("utf8mb4");

?>

```

CONCEPT mysqli creates the connection between PHP and the alumni_system database. If your XAMPP MySQL root account has a password, replace the empty password value.

9. Alumni Registration Form — index.html

Create index.html. The form is divided into Personal, Academic, Employment, Address and Account sections.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
  <title>Alumni Registration</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
  <div class="form-card">

    <div class="header">
      <h1>Alumni Registration</h1>
      <p>Join our Alumni Community</p>
    </div>

    <form action="register.php" method="POST">

      <h2>Personal Information</h2>

      <div class="form-group">
        <label for="full_name">Full Name</label>
        <input type="text" id="full_name"
          name="full_name"
          placeholder="Enter your full name" required>
      </div>

      <div class="form-group">
        <label for="email">Email Address</label>
        <input type="email" id="email"
          name="email"
          placeholder="alumni@example.com" required>
      </div>

      <div class="form-group">
        <label for="phone">Phone Number</label>
        <input type="text" id="phone"
          name="phone"
          placeholder="03XXXXXXXX" required>
      </div>

      <div class="form-group">
        <label>Gender</label>
        <div class="radio-group">
          <label><input type="radio" name="gender"
            value="Male" required> Male</label>
          <label><input type="radio" name="gender"
            value="Female"> Female</label>
        </div>
      </div>

      <div class="form-group">
        <label for="date_of_birth">Date of Birth</label>
        <input type="date" id="date_of_birth"
          name="date_of_birth" required>
      </div>

    </form>

  </div>
</div>
```

```

<h2>Academic Information</h2>

<div class="form-group">
  <label for="student_id">Student ID / Registration No.</label>
  <input type="text" id="student_id"
    name="student_id"
    placeholder="Enter student ID" required>
</div>

<div class="form-group">
  <label for="degree">Degree / Program</label>
  <select id="degree" name="degree" required>
    <option value="">Select Degree</option>
    <option>BS Computer Science</option>
    <option>BS Software Engineering</option>
    <option>BS Information Technology</option>
    <option>BS Artificial Intelligence</option>
    <option>MS Computer Science</option>
    <option>MBA</option>
  </select>
</div>

<div class="form-group">
  <label for="department">Department</label>
  <select id="department" name="department" required>
    <option value="">Select Department</option>
    <option>Computer Science</option>
    <option>Software Engineering</option>
    <option>Information Technology</option>
    <option>Business Administration</option>
    <option>Electrical Engineering</option>
  </select>
</div>

<div class="form-group">
  <label for="graduation_year">Graduation Year</label>
  <select id="graduation_year"
    name="graduation_year" required>
    <option value="">Select Year</option>
    <option>2026</option><option>2025</option>
    <option>2024</option><option>2023</option>
    <option>2022</option><option>2021</option>
    <option>2020</option><option>2019</option>
    <option>2018</option><option>2017</option>
    <option>2016</option><option>2015</option>
  </select>
</div>

<h2>Employment Information</h2>

<div class="form-group">
  <label for="current_job">Current Job / Designation</label>
  <input type="text" id="current_job"
    name="current_job"
    placeholder="e.g. Software Engineer">
</div>

<div class="form-group">
  <label for="company">Organization / Company</label>
  <input type="text" id="company"
    name="company"
    placeholder="Enter organization name">
</div>

```

```

<h2>Current Address</h2>

<div class="form-group">
  <label for="city">City</label>
  <input type="text" id="city"
    name="city" placeholder="Enter city" required>
</div>

<div class="form-group">
  <label for="country">Country</label>
  <input type="text" id="country"
    name="country" value="Pakistan" required>
</div>

<div class="form-group">
  <label for="address">Complete Address</label>
  <textarea id="address" name="address"
    rows="4"
    placeholder="Enter current address"></textarea>
</div>

<h2>Account Information</h2>

<div class="form-group">
  <label for="password">Password</label>
  <input type="password" id="password"
    name="password"
    placeholder="Create a password" required>
</div>

<button type="submit">Register as Alumni</button>

</form>
</div>
</div>

</body>
</html>

```

KEY IDEA The HTML name attribute is the bridge to PHP: name="email" is received as \$_POST['email']; name="student_id" is received as \$_POST['student_id'].

10. CSS Styling — style.css

```

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, sans-serif;
  background: #eef2f7;
  padding: 40px 20px;
}

.container {
  max-width: 850px;
  margin: auto;
}

```

```

.form-card {
    background: white;
    padding: 40px;
    border-radius: 16px;
    box-shadow: 0 10px 35px rgba(0,0,0,.10);
}

.header {
    text-align: center;
    margin-bottom: 35px;
}

.header h1 {
    font-size: 32px;
    margin-bottom: 10px;
}

.header p {
    color: #64748b;
}

h2 {
    font-size: 20px;
    margin: 30px 0 18px;
    padding-bottom: 8px;
    border-bottom: 2px solid #e2e8f0;
}

.form-group {
    margin-bottom: 20px;
}

.form-group label {
    display: block;
    margin-bottom: 8px;
    font-weight: bold;
}

input, select, textarea {
    width: 100%;
    padding: 13px;
    border: 1px solid #cbd5e1;
    border-radius: 8px;
    font-size: 15px;
}

input:focus, select:focus, textarea:focus {
    outline: none;
    border-color: #2563eb;
}

.radio-group {
    display: flex;
    gap: 25px;
}

.radio-group label {
    font-weight: normal;
}

.radio-group input {
    width: auto;
    margin-right: 5px;
}

```

```

button {
    width: 100%;
    padding: 15px;
    border: none;
    border-radius: 8px;
    background: #2563eb;
    color: white;
    font-size: 17px;
    font-weight: bold;
    cursor: pointer;
    margin-top: 15px;
}

button:hover {
    background: #1d4ed8;
}

```

11. PHP Backend — register.php

The backend validates the request, checks for duplicate email addresses, hashes the password and inserts the record with a prepared statement.

```

<?php

require_once "db.php";

if ($_SERVER["REQUEST_METHOD"] !== "POST") {
    die("Invalid request.");
}

$full_name = trim($_POST["full_name"] ?? "");
$email = trim($_POST["email"] ?? "");
$phone = trim($_POST["phone"] ?? "");
$gender = trim($_POST["gender"] ?? "");
$date_of_birth = $_POST["date_of_birth"] ?? "";
$student_id = trim($_POST["student_id"] ?? "");
$degree = trim($_POST["degree"] ?? "");
$department = trim($_POST["department"] ?? "");
$graduation_year = $_POST["graduation_year"] ?? "";
$current_job = trim($_POST["current_job"] ?? "");
$company = trim($_POST["company"] ?? "");
$city = trim($_POST["city"] ?? "");
$country = trim($_POST["country"] ?? "");
$address = trim($_POST["address"] ?? "");
$password = $_POST["password"] ?? "";

if (
    empty($full_name) || empty($email) ||
    empty($phone) || empty($gender) ||
    empty($date_of_birth) || empty($student_id) ||
    empty($degree) || empty($department) ||
    empty($graduation_year) || empty($city) ||
    empty($country) || empty($password)
) {
    die("Please fill all required fields.");
}

if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    die("Invalid email address.");
}

```

```

/* Check duplicate email */
$check = $conn->prepare(
    "SELECT id FROM alumni WHERE email = ?"
);
$check->bind_param("s", $email);
$check->execute();
$check->store_result();

if ($check->num_rows > 0) {
    die("This email address is already registered.");
}
$check->close();

/* Secure password */
$password_hash = password_hash(
    $password,
    PASSWORD_DEFAULT
);

/* Insert alumni */
$sql = "INSERT INTO alumni
(
    full_name, email, phone, gender, date_of_birth,
    student_id, degree, department, graduation_year,
    current_job, company, city, country, address, password
)
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";

$stmt = $conn->prepare($sql);

$stmt->bind_param(
    "ssssssssssss",
    $full_name, $email, $phone, $gender, $date_of_birth,
    $student_id, $degree, $department, $graduation_year,
    $current_job, $company, $city, $country, $address,
    $password_hash
);

if ($stmt->execute()) {
    echo "Registration successful. Welcome to the Alumni Community.";
} else {
    echo "Registration failed: " . $stmt->error;
}

$stmt->close();
$conn->close();

?>

```

SECURITY Passwords should never be stored as plain text. `password_hash()` creates a one-way password hash. Prepared statements also reduce SQL-injection risk.

12. Alumni Directory — alumni.php

```

<?php

require_once "db.php";

$sql = "SELECT id, full_name, email, phone,
    student_id, degree, department,
    graduation_year, current_job, company,

```

```

        city, country, created_at
    FROM alumni
    ORDER BY id DESC";

$result = $conn->query($sql);

?>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"
    content="width=device-width, initial-scale=1.0">
<title>Alumni Directory</title>
<style>
body { font-family: Arial; background:#f1f5f9; padding:30px; }
.container { max-width:1400px; margin:auto; }
.table-wrapper { overflow-x:auto; }
table { width:100%; border-collapse:collapse; background:white; }
th, td { padding:12px; border:1px solid #ddd; text-align:left; }
th { background:#2563eb; color:white; }
tr:nth-child(even) { background:#f8fafc; }
</style>
</head>

<body>
<div class="container">
<h1>Alumni Directory</h1>

<div class="table-wrapper">
<table>
<tr>
<th>ID</th><th>Name</th><th>Email</th><th>Phone</th>
<th>Student ID</th><th>Degree</th><th>Department</th>
<th>Graduation</th><th>Job</th><th>Company</th>
<th>City</th><th>Country</th><th>Registered</th>
</tr>

<?php
if ($result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        ?>
        <tr>
        <td><?= htmlspecialchars($row["id"]) ?></td>
        <td><?= htmlspecialchars($row["full_name"]) ?></td>
        <td><?= htmlspecialchars($row["email"]) ?></td>
        <td><?= htmlspecialchars($row["phone"]) ?></td>
        <td><?= htmlspecialchars($row["student_id"]) ?></td>
        <td><?= htmlspecialchars($row["degree"]) ?></td>
        <td><?= htmlspecialchars($row["department"]) ?></td>
        <td><?= htmlspecialchars($row["graduation_year"]) ?></td>
        <td><?= htmlspecialchars($row["current_job"]) ?></td>
        <td><?= htmlspecialchars($row["company"]) ?></td>
        <td><?= htmlspecialchars($row["city"]) ?></td>
        <td><?= htmlspecialchars($row["country"]) ?></td>
        <td><?= htmlspecialchars($row["created_at"]) ?></td>
        </tr>
        <?php
    }
} else {
    ?>
    <tr><td colspan="13">No alumni registered yet.</td></tr>
    <?php } ?>

```

```

</table>
</div>
</div>
</body>
</html>

<?php $conn->close(); ?>

```

OUTPUT Open <http://localhost/alumni-system/alumni.php> to see the records retrieved from MySQL.

13. Troubleshooting

Problem	Possible Solution
Apache not starting	Check whether another service is using the Apache port; inspect XAMPP logs.
MySQL not starting	Check for another MySQL/MariaDB service and inspect MySQL logs.
Database connection failed	Verify host, username, password and database name in db.php.
404 Not Found	Confirm the folder is C:\xampp\htdocs\alumni-system.
PHP code appears in browser	Use http://localhost/... rather than opening the PHP file directly.
Table does not exist	Create the alumni table inside the alumni_system database.
Duplicate email	Use a different email; email is defined as UNIQUE.

14. Exercises & Challenge Task

Add a profile photo upload field. Investigate input type=file and enctype="multipart/form-data".

15. Viva Questions

1. What is HTML and why is it used in this project?
2. What is the purpose of the form action attribute?
3. What is the difference between GET and POST?
4. Why is POST appropriate for registration?
5. What is \$_POST in PHP?
6. Why is the name attribute important in form controls?
7. What is PHP?
8. What is mysqli?
9. What is a prepared statement?
10. What does bind_param() do?
11. What does password_hash() do?
12. Why should passwords not be stored as plain text?
13. What is SQL injection?
14. What is a primary key?
15. What does AUTO_INCREMENT do?
16. What is UNIQUE?

17. What is XAMPP?
18. What is Apache?
19. What is phpMyAdmin?
20. Why must PHP files be accessed through localhost?
21. What is CRUD?
22. Why should an alumni system have an administrator?

By completing this laboratory, students have built a working component of a real-world Alumni Management System and learned the essential data flow: